

8. You are preparing AI-generated explanations for a medical summary where minor logical inconsistencies could lead to patient misunderstanding. Which reasoning path is most appropriate as a starting point?

1 / 1 point

- ☐ Direct answer for speed
- ☐ Creative brainstorming
- ☒ Step-by-step reasoning with self-critique
- ☐ Step-by-step reasoning only

✔ Nice work

High-stakes contexts justify additional verification layers. This structure reduces certain classes of reasoning error while making assumptions visible for review.

6. Even after adding step-by-step reasoning and self-critique, the model produces an answer based on incorrect factual premises. What is the most accurate conclusion?

1 / 1 point

- ☐ The model ignored instructions
- ☒ Reasoning cannot fix missing or incorrect knowledge
- ☐ Temperature should be increased
- ☐ Reasoning failed and should not be used

✔ Nice work

Structured reasoning improves logical processing but cannot supply facts the model does not know. Recognizing this boundary prevents misplaced trust in scaffolds.

7. You add self-critique to a moderately complex planning task. The responses become significantly longer, but measurable accuracy improves only slightly. What is the most responsible interpretation?

1 / 1 point

- ☐ Direct answers are always superior
- ☐ The model requires retraining
- ☐ The model is thinking harder and therefore better
- ☒ Additional structure increased cost with limited benefit

✔ Nice work

Self-critique consumes more tokens and time. If accuracy gains are marginal, the system designer must weigh cost against benefit. This reflects disciplined deployment thinking.

4. You ask an AI to generate a list of dinner ideas based on ingredients you already have. Which reasoning path is least appropriate?

1 / 1 point

- ☐ Any approach is acceptable
- ☒ Self-critique and revision
- ☐ Direct answer
- ☐ Step-by-step reasoning

✔ Nice work

Adding critique layers increases computational cost and latency without meaningful risk reduction for a casual task. Responsible system design matches reasoning depth to stakes.

5. A learner observes that a response with self-critique appears "smarter" than a direct answer. Which explanation best aligns with course concepts?

1 / 1 point

- ☒ Structured reasoning exposed more intermediate steps
- ☐ The model temporarily became more intelligent
- ☐ Additional reasoning tokens changed the model's weights
- ☐ The model accessed external knowledge

✔ Nice work

Self-critique increases the visible reasoning process and allows inconsistencies to surface. The improvement comes from structure and additional evaluation, not hidden weight changes.

2. A complex word problem produces an incorrect final answer under a direct-response approach. When step-by-step reasoning is added, the arithmetic improves, but the final conclusion is still wrong due to a faulty assumption. What does this outcome most strongly suggest?

1 / 1 point

- ☐ The model lacks sufficient intelligence
- ☒ Reasoning structure reduced some errors but not all
- ☐ Step-by-step reasoning guarantees correctness
- ☐ The problem requires retraining the model

✔ Nice work

Step-by-step reasoning improved arithmetic accuracy but did not correct the faulty assumption. This demonstrates that scaffolds reshape error distribution rather than eliminate it.

3. You deploy step-by-step reasoning for financial calculations. You notice that arithmetic errors decrease, but occasional inconsistencies remain in the final totals. What is the most appropriate next intervention?

1 / 1 point

- ☐ Retrain the model on financial data
- ☐ Increase temperature for creativity
- ☐ Remove step-by-step reasoning
- ☒ Add a self-critique or verification pass after the solution

✔ Nice work

Introducing a review stage directly targets residual inconsistencies. This reflects iterative system design: evaluate performance, then add the smallest structure needed to reduce remaining errors.



Back



Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. You are shown three AI responses to the same multi-step analytical problem.

1 / 1 point

Response A: Provides a short, confident answer with no explanation

Response B: Walks through the problem step by step

Response C: Solves the problem step by step, then reviews its own answer for errors

You are designing a workflow where small mistakes are costly and outputs must be checked before delivery. Which response structure would you deploy?

☐ Response B

☐ Response A

☒ Response C

✔ Nice work

Step-by-step reasoning followed by self-critique introduces a second layer of evaluation. This structure does not make the model smarter, but it reduces certain classes of error. Choosing this path reflects risk-aware system design.

8. You are preparing AI-generated explanations for a medical summary where minor logical inconsistencies could lead to patient misunderstanding. Which reasoning path is most appropriate as a starting point?

1 / 1 point

- ☒ Step-by-step reasoning with self-critique
- ☐ Creative brainstorming
- ☐ Direct answer for speed
- ☐ Step-by-step reasoning only

✔ Nice work

High-stakes contexts justify additional verification layers. This structure reduces certain classes of reasoning error while making assumptions visible for review.

6. Even after adding step-by-step reasoning and self-critique, the model produces an answer based on incorrect factual premises. What is the most accurate conclusion?

1 / 1 point

- ☐ Temperature should be increased
- ☒ Reasoning cannot fix missing or incorrect knowledge
- ☐ The model ignored instructions
- ☐ Reasoning failed and should not be used

 **Nice work**

Structured reasoning improves logical processing but cannot supply facts the model does not know. Recognizing this boundary prevents misplaced trust in scaffolds.

7. You add self-critique to a moderately complex planning task. The responses become significantly longer, but measurable accuracy improves only slightly. What is the most responsible interpretation?

1 / 1 point

- ☐ The model is thinking harder and therefore better
- ☒ Additional structure increased cost with limited benefit
- ☐ The model requires retraining
- ☐ Direct answers are always superior

 **Nice work**

Self-critique consumes more tokens and time. If accuracy gains are marginal, the system designer must weigh cost against benefit. This reflects disciplined deployment thinking.

4. You ask an AI to generate a list of dinner ideas based on ingredients you already have. Which reasoning path is least appropriate?

0 / 1 point

- ☐ Direct answer
- ☐ Any approach is acceptable
- ☒ Step-by-step reasoning
- ☐ Self-critique and revision

⊗ Try again

While unnecessary, step-by-step reasoning would not fundamentally harm the task. It may add verbosity but still produce usable results.

5. A learner observes that a response with self-critique appears "smarter" than a direct answer. Which explanation best aligns with course concepts?

1 / 1 point

- ☐ The model temporarily became more intelligent
- ☐ The model accessed external knowledge
- ☒ Structured reasoning exposed more intermediate steps
- ☐ Additional reasoning tokens changed the model's weights

✓ Nice work

Self-critique increases the visible reasoning process and allows inconsistencies to surface. The improvement comes from structure and additional evaluation, not hidden weight changes.

2. A complex word problem produces an incorrect final answer under a direct-response approach. When step-by-step reasoning is added, the arithmetic improves, but the final conclusion is still wrong due to a faulty assumption. What does this outcome most strongly suggest?

1 / 1 point

- ☐ The problem requires retraining the model
- ☐ The model lacks sufficient intelligence
- ☒ Reasoning structure reduced some errors but not all
- ☐ Step-by-step reasoning guarantees correctness

✔ Nice work

Step-by-step reasoning improved arithmetic accuracy but did not correct the faulty assumption. This demonstrates that scaffolds reshape error distribution rather than eliminate it.

3. You deploy step-by-step reasoning for financial calculations. You notice that arithmetic errors decrease, but occasional inconsistencies remain in the final totals. What is the most appropriate next intervention?

1 / 1 point

- ☐ Increase temperature for creativity
- ☐ Retrain the model on financial data
- ☒ Add a self-critique or verification pass after the solution
- ☐ Remove step-by-step reasoning

✔ Nice work

Introducing a review stage directly targets residual inconsistencies. This reflects iterative system design: evaluate performance, then add the smallest structure needed to reduce remaining errors.



Back

One Problem, Three Reasoning Paths

Graded Assignment • 30 min



Due Jun 8, 11:59 PM PDT

Your grade: **87.50%**

Your latest: **87.50%** • Your highest: **87.50%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. You are shown three AI responses to the same multi-step analytical problem.

1 / 1 point

Response A: Provides a short, confident answer with no explanation

Response B: Walks through the problem step by step

Response C: Solves the problem step by step, then reviews its own answer for errors

You are designing a workflow where small mistakes are costly and outputs must be checked before delivery. Which response structure would you deploy?

☐ Response B

☐ Response A

☒ Response C



Nice work

Step-by-step reasoning followed by self-critique introduces a second layer of evaluation. This structure does not make the model smarter, but it reduces certain classes of error. Choosing this path reflects risk-aware system design.